

CYBER INCIDENT RESPONSE & SECURITY BEST PRACTICES REPORT

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1. INTRODUCTION

Cyber incidents can lead to data breaches, operational disruption, financial losses, and reputational damage. Incident response helps organizations manage cyber attacks effectively and reduce their impact.

1. OBJECTIVE

- Understand the incident response lifecycle.
 - Analyze a real-world cyber security incident.
 - Develop a basic incident response plan.
 - Learn security best practices.
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1. INCIDENT RESPONSE LIFECYCLE

Identification

The organization detects and confirms a security incident through monitoring systems, alerts, and security investigations.

Containment

Immediate actions are taken to prevent the incident from spreading and causing additional damage.

Eradication

The root cause of the incident is removed, including malware, unauthorized accounts, or exploited vulnerabilities.

Recovery

Systems and services are restored to normal operation after ensuring they are secure and stable.

Lessons Learned

The incident is reviewed to improve future security measures and response procedures.

1. CASE STUDY – WANNACRY RANSOMWARE ATTACK

The WannaCry ransomware attack occurred in 2017 and affected hundreds of thousands of computers worldwide.

Cause:

Exploitation of an unpatched Windows vulnerability.

Impact:

- Service disruption
- Financial losses
- Data inaccessibility

Lessons Learned:

- Regular software updates
 - Reliable backup systems
 - Effective incident response planning
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1. HYPOTHETICAL INCIDENT RESPONSE PLAN

Scenario:

An employee clicks a phishing email attachment that installs malware.

Response Actions:

1. Identify affected devices.
 2. Disconnect infected systems from the network.
 3. Remove malicious software.
 4. Restore systems using backups.
 5. Monitor for suspicious activity.
 6. Conduct awareness training for employees.
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1. SECURITY BEST PRACTICES CHECKLIST

- ✓ Use Strong Passwords
- ✓ Enable Multi-Factor Authentication
- ✓ Update Software Regularly
- ✓ Maintain Secure Backups

✓ Conduct Security Awareness Training

✓ Monitor Logs and Alerts

✓ Restrict Access Based on Roles

✓ Use Antivirus Protection

1. CHALLENGES FACED

One of the main challenges during this task was understanding how each phase of the incident response lifecycle works together during a real cyber attack. Analyzing a real-world incident helped me understand the importance of preparation, quick decision-making, and recovery planning.

1. CONCLUSION

This task improved my understanding of cyber incident response, cyber attack management, and organizational security practices. It highlighted the importance of preparedness, timely response, and continuous improvement in cyber security operations.

1. SCREENSHOT REFERENCES

Figure 1 – Incident Response Lifecycle Diagram

Figure 2 – Gmail Phishing Warning